

YASHRAJ AGAWANE

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Professional Summary

Software Developer and Machine Learning Enthusiast with strong foundations in Java, Python, C++, and Data Structures. Experienced in building AI-powered applications, full-stack web development, and real-time systems using modern frameworks, REST APIs, and machine learning techniques. Seeking opportunities to apply problem-solving and software engineering skills in impactful projects.

Skills

Programming: Java, Python, C++, C, SQL

Web Technologies: Next.js, React, HTML, CSS, JSP, .NET

Machine Learning: Scikit-learn, XGBoost, Data Preprocessing, Feature Engineering, Model Evaluation

Tools: Git, GitHub, Streamlit, Render, Vercel, VS Code

Concepts: Data Structures and Algorithms, OOP, DBMS, REST APIs, Computer Vision, Operating Systems

Projects

AI Fitness Coach (Full-Stack AI Application) | *Next.js, React, TailwindCSS, Gemini API* 📄 **Jan 2026 – Mar 2026**

- **Engineered** a full-stack AI fitness and nutrition application, integrating the **Gemini LLM API** to generate highly personalized workout and diet plans, serving **100+ simulated users**.
- **Implemented** real-time streaming responses and multi-session chat utilizing **React** and **localStorage** for robust state management, improving perceived load times by **40%**.
- **Architected** scalable **RESTful APIs** and server-side rendered (SSR) dynamic UI components utilizing **Next.js** and **TailwindCSS**, ensuring seamless data flow across the platform.

Driver Drowsiness Detection System | *Python, Flask, OpenCV, Dlib, Computer Vision* 📄 **Nov 2025 – Jan 2026**

- **Developed** a real-time computer vision monitoring system leveraging **OpenCV** and **Dlib** to compute Eye Aspect Ratio (EAR) via facial landmark detection for robust fatigue identification.
- **Integrated** a **WebRTC** browser interface with a **Python Flask** backend, enabling continuous, low-latency video frame processing at **30+ FPS** and instantaneous alert generation.
- **Deployed** a live interactive dashboard to visualize drowsiness metrics and trigger automated safety alerts, successfully demonstrating a **98% detection rate** in simulated fatigue scenarios.

Multi-Disease Prediction System | *Python, Machine Learning, Scikit-learn, XGBoost* 📄 **Oct 2025 – Nov 2025**

- **Built** an end-to-end predictive ML system for early diagnosis of multiple chronic diseases (diabetes, heart disease, cancer) by processing and analyzing diverse healthcare datasets.
- **Designed** a robust ML pipeline encompassing rigorous data preprocessing, feature engineering, and predictive modeling using **XGBoost**, **Random Forest**, and **SVM** algorithms.
- **Optimized** model performance via systematic feature selection and **GridSearchCV** hyperparameter tuning, elevating the baseline prediction accuracy to an industry-standard **95%** while minimizing false negatives.

Awards & Certifications

- **Artificial Intelligence Bootcamp – C-DAC Pune (FutureSkills PRIME):** Machine Learning, Deep Learning, Computer Vision, NLP; scored **86/100 (Very Good)**.
- **Programming in Python – NPTEL (2024):** Top 1%, scored **87%**; Python, Data Structures.
- **Git Training – IIT Bombay Spoken Tutorial:** Version Control using Git and GitHub; scored **80%**.
- **Machine Learning Using Python – Simplilearn SkillUp (Oct 2025):** Data Preprocessing, Supervised Learning, Model Training, Machine Learning with Python.

Education

B.Tech. Information Technology

Vidya Pratishthan's Kamalnayan Bajaj Institute of Engineering and Technology, Baramati

Aug 2023 – May 2027

CGPA: 8.55 / 10